

Array Coding Patterns (Java Focus)

1. Traversal Pattern

Used for printing, sum, max/min, counting.

Template:

```
for(int i = 0; i < arr.length; i++) {  
    // use arr[i]  
}
```

2. Two Pointer Pattern

Used in sorted arrays, reverse array, pair sum.

Template:

```
int left = 0;  
int right = arr.length - 1;  
while(left < right) {  
    left++;  
    right--;  
}
```

3. Sliding Window Pattern

Used for subarray problems and maximum sum of k elements.

Template (Fixed Size):

```
int windowSum = 0;  
for(int i = 0; i < k; i++) {  
    windowSum += arr[i];  
}  
for(int i = k; i < arr.length; i++) {  
    windowSum += arr[i] - arr[i - k];  
}
```

4. Prefix Sum Pattern

Used for range sum queries.

Template:

```
prefix[0] = arr[0];  
for(int i = 1; i < arr.length; i++) {  
    prefix[i] = prefix[i - 1] + arr[i];  
}
```

5. Kadane's Algorithm

Used for maximum subarray sum.

Template:

```
int maxSum = arr[0];  
int currentSum = arr[0];  
for(int i = 1; i < arr.length; i++) {  
    currentSum = Math.max(arr[i], currentSum + arr[i]);  
    maxSum = Math.max(maxSum, currentSum);  
}
```

6. Frequency / Hashing Pattern

Used to count occurrences or detect duplicates.

Template:

```
HashMap<Integer, Integer> map = new HashMap<>();
```

7. Sorting Based Pattern

Used in interval problems and grouping problems.

Template:

```
Arrays.sort(arr);
```

8. Cyclic Sort Pattern

Used when numbers are from 1 to N.

Helpful for finding missing or duplicate numbers.